「Def. Let S be a regular surface and T_pS be the tangent plane at $p \in S$.
A quadratic form

$$I_p : T_pS \to \mathbb{R} : w \mapsto \langle w, w \rangle_p \geq 0$$

is called the first fundamental form of S at p.

「Computation of the fundamental form.
Given $p \in S$, let $X : U \to S$ be a parametrization of S containing p.
Let $w \in T_pS$ be given. Take a curve

$$\alpha : (-\varepsilon, \varepsilon) \to X[U] : t \mapsto X(u(t), v(t))$$ such that $\alpha(0) = p$ and $\alpha'(0) = w$.

Now,

$$\begin{aligned} I_p(w) = I_p(\alpha'(0)) &= \langle \alpha'(0), \alpha'(0) \rangle_p \\ &= \langle X_u u' + X_v v', X_u u' + X_v v' \rangle \\ &= \langle X_u, X_u \rangle \cdot (u')^2 + 2 \cdot \langle X_u, X_v \rangle \cdot u'v' + \langle X_v, X_v \rangle \cdot (v')^2. \quad (t=0). \end{aligned}$$

In this sence, define

$$E(u_0, v_0) = \langle X_u, X_u \rangle_p, \quad F(u_0, v_0) = \langle X_u, X_v \rangle_p, \quad G(u_0, v_0) = \langle X_v, X_v \rangle_p.$$

「Def. A length of parametrized curve $\alpha : I \to S$ is

$$S(t) = \int_0^t |\alpha'(r)| \, dr = \int_0^t \sqrt{I(\alpha'(r))} \, dr = \int_0^t \sqrt{E \cdot (u')^2 + 2F \cdot u'v' + G(v')^2} \, dr$$

「Def. Let $R \subset S$ be a bounded set contained a image of $X : U \to S$,
A area of R is:

$$A(R) = \iint_{X^{-1}[R]} |X_u \times X_v| \, du dv.$$

「Examples,
Let $S=\{(x,y,z)\mid x^2+y^2=1\}$ be the Sylinder.
Consider a Parametrization of S,

$$X: U \to \mathbb{R}^3 : (u,v) \mapsto (\cos u, \sin u, v)$$

where $U=\{(u,v)\in\mathbb{R}^2 \mid 0<u<2\pi, -\infty<v<\infty\}$,
Simply, $X_u=(-\sin u, \cos u, 0)$ and $X_v=(0,0,1)$.
So, $E=1$, $F=0$, $G=1$.

「Helicoid.
Let $S=\{(r\cdot\cos\theta, r\cdot\sin\theta, a$

Exercises. Sec 2.5.

1. Evaluate the first-fundamental form of the following:

a). $X(u,v) = (a \cdot \sin u \cdot \cos v,\ b \cdot \sin u \cdot \sin v,\ c \cdot \cos u)$

b). $X(u,v) = (a \cdot u \cos u,\ b \cdot u \sin v,\ u^2)$

Proof.

a), $X_u = (a \cos u \cdot \cos v,\ b \cos u \cdot \sin v,\ -c \cdot \sin u)$,

$X_v = (-a \sin u \cdot \sin v,\ b \cdot \sin u \cdot \cos v,\ 0)$.

So, $E = \langle X_u, X_u \rangle = a^2 \cdot \cos^2 u \cdot \cos^2 v + b^2 \cos^2 u \cdot \sin^2 v + c^2 \cdot \sin^2 u$

$= (a^2 \cos^2 v + b^2 \sin^2 v) \cos^2 u + c^2 \cdot \sin^2 u$

$F = \langle X_u, X_v \rangle = -a^2 \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v + b^2 \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v$

$= (-a^2 + b^2) \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v$.

$G = \langle X_v, X_v \rangle = a^2 \cdot \sin^2 u \cdot \sin^2 v + b^2 \cdot \sin^2 v \cdot \cos^2 v$.

First Fundamental form of the Monge patch.

Given a differentiable map $f: \mathbb{R}^2 \to \mathbb{R}$, consider the Monge patch

$$X : U \to \mathbb{R}^3 : (u, v) \mapsto (u, v, f(u,v)).$$

Then, the partial derivative is

$$X_u = (1, 0, f_u) \text{ and } X_v = (0, 1, f_v)$$

So, the first-fundamental form is

$$E = \langle X_u, X_u \rangle = 1 + (f_u)^2, \quad F = \langle X_u, X_v \rangle = f_u \cdot f_v, \quad G = \langle X_v, X_v \rangle = 1 + (f_v)^2.$$

Moreover, $\det\begin{pmatrix} E & F \\ F & G \end{pmatrix} = 1 + (f_u)^2 + (f_v)^2 \geq 1 > 0.$

Notation

Denote the Coordinate Patch as

$$X: U \to \mathbb{R}^3 ; (u^1, u^2) \mapsto (f^1, f^2, f^3)$$

「Second fundamental form.

「Def. Let $X : \mathcal{U} \to \mathbb{R}^3$ be a Coordinate Patch.
The second fundamental form of X is:

$$L_{ij} \overset{\text{def}}{=} \langle X_{ij}, n \rangle \quad (1 \leq i, j \leq 2)$$

And the Christoffel Symbols is

$$\Gamma_{ij}^{k} \overset{\text{def}}{=} \sum_{l=1}^{2} \langle X_{ij}, X_l \rangle \cdot g^{lk}.$$

Where $X_{ij} := \dfrac{\partial^2 X}{\partial u^i \partial u^j}$ and g^{lk} is (k, l) entry of $\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}^{-1}$.

$$n = \frac{X_1 \times X_2}{\|X_1 \times X_2\|}. \qquad = \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}.$$

Normal and geodesic Curvature.

Def Let $X : \mathcal{U} \to \mathbb{R}^3$ be a patch and

$$\gamma : I \to \mathbb{R}^3 : s \mapsto X(\gamma^1(s), \gamma^2(s))$$

be a unit speed curve whose image is contained in the patch.
Define the Normal Curvature

$$\kappa_n(s) = \langle \gamma''(s), n(s) \rangle$$

and the Geodesic Curvature

$$\kappa_g(s) = \langle \gamma''(s), n(s) \times T(s) \rangle$$

So that $\kappa(s) \cdot N(s) = T'(s) = \gamma''(s) = \kappa_n(s) \cdot n(s) + \kappa_g(s) \cdot n(s) \times T(s)$.

And $\kappa^2 = \kappa_n^2 + \kappa_g^2$

Normal of Monge patch.

$$\begin{matrix} 1 & 0 & f_u \\ 0 & 1 & f_v \end{matrix}$$

Let $X : U \to \mathbb{R}^3 : (u, v) \mapsto (u, v, f(u, v))$

Then,

$$n = \frac{X_u \times X_v}{\| X_u \times X_v \|} = \frac{(-f_u, -f_v, 1)}{\sqrt{1 + f_u^2 + f_v^2}}$$

Example.

Consider the upper hemisphere

$$X : \mathbb{D} \to \mathbb{R}^3 : (u, v) \mapsto (u, v, \sqrt{1-u^2-v^2})$$

Let a unit speed curve in X

$$\gamma : (-\tfrac{\pi}{2}, \tfrac{\pi}{2}) \to X[\mathbb{D}] : t \mapsto (\sin t, 0, \cos t).$$

The Frenet frame is

$$f_u = \frac{-2u}{2 \cdot f} = -\frac{1}{f}, \quad f_v = -\frac{1}{f}.$$

$T(t) = (\cos t, 0, -\sin t)$ $\qquad \kappa(s) = 1$

$N(t) = (-\sin t, 0, -\cos t)$ $\qquad \tau(s) = 0.$

$B(t) = (0, 1, 0).$

$$f_u = -u/f, \quad f_v = -v/f$$

And the normal

$$n(t) = (u/f, v/f, 1) \Big/ \underbrace{\sqrt{1 + \frac{u^2+v^2}{f^2}}}_{=f}$$

$$1 + \frac{u^2+v^2}{f^2} = \frac{1-u^2-v^2}{f^2}$$

$$= (u, v, f).$$

The normal curvature

$$\kappa_n =$$